AI-Based Fraud Detection in Government Schemes

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Team Number: FXH26-P-BDDS-029

Event: FUSIONX Hackathon

Date: 14 April 2026

GitHub Repository Link: https://github.com/saniyashariff61281-sys/-fraud-detection-project.git

Introduction

Government welfare schemes play a crucial role in supporting citizens. However, these systems often face issues such as fraudulent beneficiaries, duplicate identities, and abnormal transaction activities, leading to financial losses and reduced transparency.

Problem Statement

Manual monitoring of large-scale beneficiary data is inefficient and prone to errors. There is a need for an automated system that can identify fraudulent patterns in government scheme data.

Proposed Solution

We propose an **AI-based Fraud Detection System** that uses machine learning techniques to identify suspicious beneficiaries based on abnormal transaction patterns.

Objectives

- Detect fraudulent beneficiaries in government schemes
- Identify abnormal transaction patterns
- Improve transparency and reduce financial losses
- Automate fraud detection using machine learning

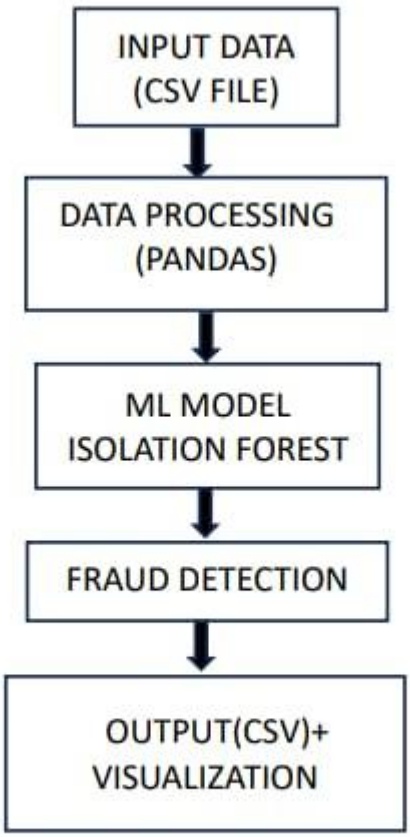
System Architecture

The system follows a data pipeline approach:

Data Input → Data Processing → Machine Learning Model → Fraud Detection → Output

- **Data Input:** CSV dataset containing user details
- **Processing:** Data cleaning and feature selection using Pandas
- **Model:** Isolation Forest algorithm for anomaly detection

- **Output:** Fraud classification (0 = Normal, 1 = Fraud)



Methodology

Step 1: Data Collection

A dataset containing user transaction details is used.

Step 2: Data Preprocessing

Relevant features such as amount, number of transactions, and age are selected.

Step 3: Model Implementation

Isolation Forest, an unsupervised machine learning algorithm, is used to detect anomalies.

Step 4: Fraud Detection

The model predicts suspicious users based on unusual patterns.

Step 5: Output Generation

Results are stored in a CSV file with fraud labels.

Tools and Technologies Used

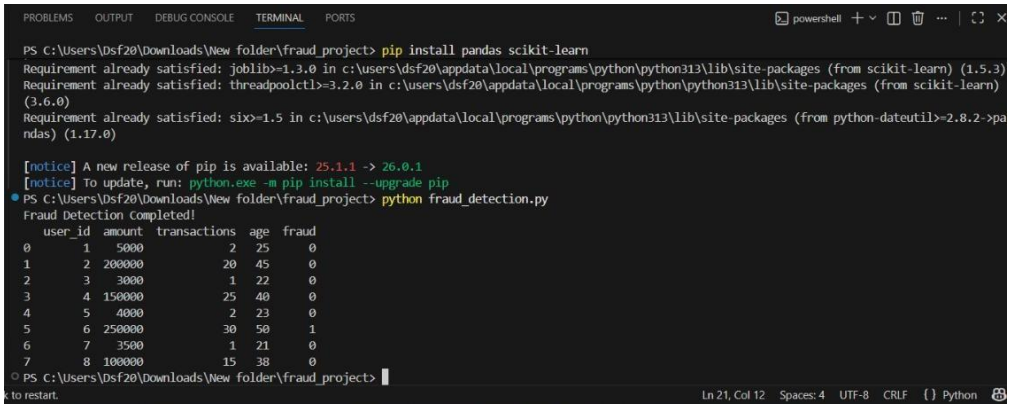
- **Programming Language:** Python
- **Libraries:** Pandas, Scikit-learn
- **Algorithm:** Isolation Forest
- **Platform:** VS Code

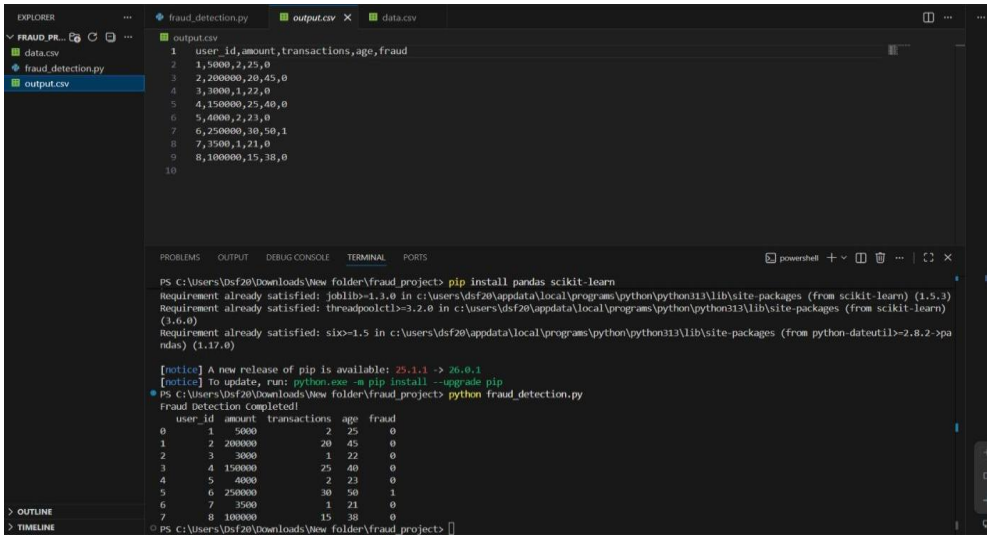
Results and Output

The system successfully identifies fraudulent users based on abnormal transaction behavior.

- **0 → Normal User**
- **1 → Fraud User**

The output is stored in output.csv, where suspicious users are clearly highlighted.





Applications

- Direct Benefit Transfer (DBT) systems
- Scholarship fraud detection

- Subsidy distribution systems
- Government welfare schemes

Advantages

- Fast and automated fraud detection
- Scalable for large datasets
- Reduces manual effort
- Improves accuracy and transparency

Limitations

- Uses sample dataset (not real-time data)
- Accuracy depends on data quality
- Limited features used in current model

Future Enhancements

- Integration with real-time government databases
- Use of advanced AI models (Deep Learning)
- Dashboard visualization using Streamlit
- Aadhaar-based duplicate detection

Conclusion

The proposed system demonstrates how machine learning can effectively detect fraudulent activities in government schemes. It provides a scalable and efficient solution to improve transparency and reduce financial losses.

References

- Scikit-learn Documentation
- Pandas Documentation
- Research papers on fraud detection systems